

New Wheels Project

Introduction to SQL

Business Context

New-Wheels is a US-based digital platform for pre-owned vehicle resale, managing everything from vehicle listings and order placement to third-party logistics (Air/Ground) and post-delivery customer feedback. Transactions are tracked via a MySQL database (vehdb) consisting of four core tables: `customer_t`, `order_t`, `product_t`, and `shipper_t`. Revenue is generated per order, factoring in vehicle price, applied discounts, and shipping modes across various vehicle makes, models, and price bands.

However, throughout 2018, New-Wheels suffered an undiagnosed quarter-on-quarter decline in both order volumes and customer satisfaction. Prior reporting initiatives failed to diagnose this decline because the company lacked a designated reporting owner, leaving no individual analyst accountable. In addition, the process collapsed under manual CSV summaries, inconsistent table joins, and a lack of standardized metrics. This operational failure peaked when the CEO received two conflicting versions of the same month's data showing a 12% revenue discrepancy. The manual reporting was abandoned, leaving leadership to make critical business decisions without a reliable, consistent view of performance.

Objective

The objective is to deliver a reproducible, data-driven business report covering Q1 to Q4 2018 that will standardize data definitions to resolve previous reporting failures and explicitly analyze four key ingredients:

1. **Net Revenue and Order Trends** (calculating net price as $\text{vehicle_price} \times (1 - \text{discount})$ to pinpoint where the sales decline began)
2. **Product Performance** (ranking top and bottom vehicle models by revenue and satisfaction)
3. **Customer Feedback Analysis** (converting text ratings into a trackable 1–5 numeric index), and
4. **Shipper Performance** (assessing logistics reliability against customer scores).

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
select
    count(distinct customer_id) as customers,
    ct.state
from customer_t ct
join order_t ot
    using(customer_id)
group by ct.state
order by count(distinct customer_id) desc
```

Output:

 Query 1

Query:

```
select
    count(distinct customer_id) as customers,
    ct.state
from customer_t ct
join order_t ot
    using(customer_id)
group by ct.state
order by count(distinct customer_id) desc
```

Output:

Showing first 10 rows out of 49 rows

customers	state
97	Texas
97	California
86	Florida
69	New York
35	District of Columbia
33	Ohio
33	Colorado
29	Alabama

Observations and Insights:

- The four most populous states in the US have the most number of orders, which is not surprising.
- The high number of orders in DC is surprising since it is a metroplex rather than a state.
- Maine, Vermont, Wyoming, Mississippi, North Dakota, Montana, and New Hampshire all have fewer than five orders. To increase profits, the company may want to limit sales to certain states (remove their presence from those states).

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
select
    pt.vehicle_maker,
    sum(ot.quantity) as total_units,
    round((sum(ot.quantity)*100.0/(select sum(quantity) from order_t)),2) as percent_market
from order_t ot
join product_t pt
    on ot.product_id = pt.product_id
group by pt.vehicle_maker
    order by total_units desc
limit 5;
```

Output:

Query 3

Query:

```
select
    pt.vehicle_maker,
    sum(ot.quantity) as total_units,
    round((sum(ot.quantity)*100.0/(select sum(quantity) from order_t)),2) as percent_market
from order_t ot
join product_t pt
    on ot.product_id = pt.product_id
group by pt.vehicle_maker
    order by total_units desc
limit 5
```

Output:

Showing 5 rows

vehicle_maker	total_units	percent_market
Chevrolet	125	8.27
Ford	97	6.42
Toyota	78	5.16
Pontiac	75	4.96
Dodge	72	4.76

Observations and Insights:

- With the exception of Toyota, American automakers dominate the top five vehicle makers.
- These vehicles comprise almost a third of the market (29.57%).

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
select
    state,
    vehicle_maker
from (
select
    ct.state,
    pt.vehicle_maker,
    count(distinct ct.customer_id) as total_customers,
    rank() over (partition by ct.state
        order by count(distinct ct.customer_id) desc) as rnk
from customer_t ct
join order_t ot
    using(customer_id)
join product_t pt
    using(product_id)
group by ct.state, pt.vehicle_maker) as state_ranks
where rnk = 1;
```

Output:

✓ Query 3

Query:

```
select
  state,
  vehicle_maker
from (
  select
    ct.state,
    pt.vehicle_maker,
    count(distinct ct.customer_id) as total_customers,
    rank() over (partition by ct.state
      order by count(distinct ct.customer_id) desc) as rnk
  from customer_t ct
  join order_t ot
    using(customer_id)
  join product_t pt
    using(product_id)
  group by ct.state, pt.vehicle_maker) as state_ranks
where rnk = 1
```

Output:

Showing first 10 rows out of 143 rows

state	vehicle_maker
Alabama	Dodge
Alaska	Chevrolet
Arizona	Pontiac
Arizona	Cadillac
Arkansas	Volkswagen

Observations and Insights:

- For this output, not much can be gleaned by me due to the limitations of the SQL Playground. Many of the states with few orders (such as Arkansas) have multiple ties for rank 1. Thus, I can't see which particular car is the most popular in this query alone.
- I ran a slightly different query (shown below) where I was able to see the max number of vehicle orders per state in states that had a clear winner (no ties for first).
- With the exception of Toyotas being the most popular car in Florida (7 sold) and Mazdas being the most popular in Indiana (4), US automakers dominated the rest of the top 10.
- This strongly suggests that the business may want to shift its focus to US automakers.

Code:

```
SELECT
    state,
    MIN(vehicle_maker) AS vehicle_maker,
    MAX(total_customers) AS total_customers
FROM (
    SELECT
        ct.state,
        pt.vehicle_maker,
        COUNT(DISTINCT ct.customer_id) AS total_customers,
        RANK() OVER (
            PARTITION BY ct.state
            ORDER BY COUNT(DISTINCT ct.customer_id) DESC
        ) AS rnk
    FROM customer_t ct
    JOIN order_t ot USING (customer_id)
    JOIN product_t pt USING (product_id)
    GROUP BY ct.state, pt.vehicle_maker
) AS state_ranks
WHERE rnk = 1
GROUP BY state
HAVING COUNT(*) = 1
order by total_customers desc;
```

order by total_customers desc

Output:

Showing first 10 rows out of 27 rows

state	vehicle_maker	total_customers
Texas	Chevrolet	9
Florida	Toyota	7
Ohio	Chevrolet	6
Washington	Chevrolet	5
Virginia	Ford	5
Maryland	Ford	5
Colorado	Chevrolet	5
Alabama	Dodge	5
Missouri	Chevrolet	4
Indiana	Mazda	4

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
select
    quarter_number,
    round(avg(feedback_rating),3) as avg_rating,
    round(avg(avg(feedback_rating)) over(),3) as overall_avg,
    max(feedback_rating) as max_rating,
    sum(case when feedback_rating = 5 then 1 else 0 end) as max_count,
    min(feedback_rating) as min_rating,
    sum(case when feedback_rating = 1 then 1 else 0 end) as min_count
from (
    select
        quarter_number,
        case
            when customer_feedback = 'Very Bad' then 1
            when customer_feedback = 'Bad' then 2
            when customer_feedback = 'Okay' then 3
            when customer_feedback = 'Good' then 4
            when customer_feedback = 'Very Good' then 5
        end as feedback_rating
    from order_t)
group by quarter_number
order by quarter_number asc;
```

Output:

✓ Query 5

Query:

```
select
  quarter_number,
  round(avg(feedback_rating),3) as avg_rating,
  round(avg(avg(feedback_rating)) over(),3) as overall_avg,
  max(feedback_rating) as max_rating,
  sum(case when feedback_rating = 5 then 1 else 0 end) as max_count,
  min(feedback_rating) as min_rating,
  sum(case when feedback_rating = 1 then 1 else 0 end) as min_count
from (
  select
    quarter_number,
    case
      when customer_feedback = 'Very Bad' then 1
      when customer_feedback = 'Bad' then 2
      when customer_feedback = 'Okay' then 3
      when customer_feedback = 'Good' then 4
      when customer_feedback = 'Very Good' then 5
    end as feedback_rating
  from order_t)
group by quarter_number
order by quarter_number asc
```

Output:

Showing 4 rows

quarter_number	avg_rating	overall_avg	max_rating	max_count	min_rating	min_count
1	3.555	3.066	5	93	1	34
2	3.355	3.066	5	75	1	39
3	2.956	3.066	5	38	1	41
4	2.397	3.066	5	20	1	61

Observations and Insights:

- The average customer rating fell consistently over each of the quarters for this year.
- The first three months showed a steady, but unacceptable downward trend. Quarter 4 showed a distinct drop in customer ratings.
- Possibly more disturbing is the drop in the number of 5 (Very Good) ratings given to the company from Q2 to Q3. The number of 1s (Very Bad) also saw a sharp spike in Q4. Clearly something went drastically wrong near the halfway point of the year and it wasn't caught.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
select
    quarter_number,
    round(sum(case when customer_feedback = 'Very Bad' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Very_Bad,
    round(sum(case when customer_feedback = 'Bad' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Bad,
    round(sum(case when customer_feedback = 'Okay' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Okay,
    round(sum(case when customer_feedback = 'Good' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Good,
    round(sum(case when customer_feedback = 'Very Good' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Very_Good
from order_t
group by quarter_number;
```

Output:

Query 6

Query:

```
select
    quarter_number,
    round(sum(case when customer_feedback = 'Very Bad' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Very_Bad,
    round(sum(case when customer_feedback = 'Bad' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Bad,
    round(sum(case when customer_feedback = 'Okay' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Okay,
    round(sum(case when customer_feedback = 'Good' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Good,
    round(sum(case when customer_feedback = 'Very Good' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Very_Good
from order_t
group by quarter_number
```

Output:

Showing 4 rows

quarter_number	Very_Bad	Bad	Okay	Good	Very_Good
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Observations and Insights:

- As seen in the previous query, there is an obvious drop in customer satisfaction at the high levels (Good and Very Good) from Q1 to Q4.
- Okay ratings remain fairly stable (which is not great from a business perspective).

- Negative ratings (Bad and Very bad) climb dramatically from Q1 to Q4. In Q1 bad ratings accounted for approximately 22% of responses (in and of itself this is high). By Q4, negative ratings account for almost 60% of all responses.
- The biggest jump is from Q3 to Q4 for strong negatives.
- The biggest drop for strong positives was Q2 to Q3.
- Again, this is highly indicative of a major failing on the part of the business in that time frame.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
select
    quarter_number as quarter,
    count(order_id) as total_orders,
    sum(quantity) as total_quant
from order_t
group by quarter_number;
```

Output:

 Query 7

Query:

```
select
    quarter_number as quarter,
    count(order_id) as total_orders,
    sum(quantity) as total_quant
from order_t
group by quarter_number
```

Output:

Showing 4 rows

quarter	total_orders	total_quant
1	310	464
2	262	400
3	229	342
4	199	306

Observations and Insights:

- I wasn't sure about the focus of the question exactly, so I pulled both the number of orders (as explicitly stated) and the volume (total_quant) of the orders for this question.
- Total orders fell by 111 from Q1 to Q4. The quantity of purchases fell by 158 vehicles from Q1 to Q4.
- Unlike the drop in customer service ratings, the drop in orders was fairly linear across quarters. This is true for both total_orders and total_quant.
- Compared with the above queries, it appears that the company began to develop a bad reputation toward the beginning of this calendar year that only accelerated, resulting in much lower demand.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
select
    quarter_number,
    net_revenue as current_q_revenue,
    lag(net_revenue) over (order by quarter_number) as previous_q_revenue,
    round((net_revenue - lag(net_revenue) over (order by quarter_number))
        *100.0 / lag(net_revenue) over (order by quarter_number), 2) as percent_change
from (
select
    quarter_number,
    round(sum((vehicle_price * (1-discount)) * quantity), 2) as net_revenue
from order_t
group by quarter_number
order by quarter_number asc);
```

Output:

Query 8

Query:

```
select
    quarter_number,
    net_revenue as current_q_revenue,
    lag(net_revenue) over (order by quarter_number) as previous_q_revenue,
    round((net_revenue - lag(net_revenue) over (order by quarter_number))
        *100.0 / lag(net_revenue) over (order by quarter_number), 2) as percent_change
from (
select
    quarter_number,
    round(sum((vehicle_price * (1-discount)) * quantity), 2) as net_revenue
from order_t
group by quarter_number
order by quarter_number asc)
```

Output:

Showing 4 rows

quarter_number	current_q_revenue	previous_q_revenue	percent_change
1	18032549.9		
2	13122995.76	18032549.9	-27.23
3	8882298.84	13122995.76	-32.32
4	8573149.28	8882298.84	-3.48

Observations and Insights:

- As expected from the drop in total orders and order quantity, there is a decreasing in quarter over quarter revenue.
- This decrease is consistent for the first part of the year. The decrease slowed down from Q3 to Q4.

- While the customer satisfaction score did not plummet until Q3, there was clearly a shift in the perception of the company: either with its quality or its reputation that caused fewer people to use its services.
- This substantial loss in revenue must be addressed before the company collapses.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
select
    quarter_number,
    total_orders,
    round((total_orders - lag(total_orders) over(order by quarter_number))
        *100.0 / lag(total_orders) over(order by quarter_number), 2) as perc_chg_orders,
    net_revenue,
    round((net_revenue - lag(net_revenue) over (order by quarter_number))
        *100.0 / lag(net_revenue) over(order by quarter_number), 2) as perc_chg_rev
from (
select
    quarter_number,
    count(order_id) as total_orders,
    round(sum((vehicle_price * (1-discount)) * quantity), 2) as net_revenue
from order_t
group by quarter_number
order by quarter_number asc);
```

Output:


Query 9

Query:

```
select
    quarter_number,
    total_orders,
    round((total_orders - lag(total_orders) over(order by quarter_number))
        *100.0 / lag(total_orders) over(order by quarter_number), 2) as perc_chg_orders,
    net_revenue,
    round((net_revenue - lag(net_revenue) over (order by quarter_number))
        *100.0 / lag(net_revenue) over(order by quarter_number), 2) as perc_chg_rev
from (
select
    quarter_number,
    count(order_id) as total_orders,
    round(sum((vehicle_price * (1-discount)) * quantity), 2) as net_revenue
from order_t
group by quarter_number
order by quarter_number asc)
```

Output:

Showing 4 rows

quarter_number	total_orders	perc_chg_orders	net_revenue	perc_chg_rev
1	310		18032549.9	
2	262	-15.48	13122995.76	-27.23
3	229	-12.6	8882298.84	-32.32
4	199	-13.1	8573149.28	-3.48

Observations and Insights:

- The side-by-side view in the query above shows a disturbing and dangerous business trend.
- Quarter over quarter, the number of total orders dropped by more than 12% points each quarter.
- The same is true of revenue. Ultimately the loss of 111 orders led to a nearly \$10MM decrease in revenue.
- For the year, there was a loss of nearly 33% of orders. This is more than a 50% loss in revenue over the entire year.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
select
  ct.credit_card_type,
  round(avg(ot.discount),3) as average_discount,
  max(ot.discount) as max_discount,
  min(ot.discount) as min_discount
from customer_t ct
join order_t ot
  using(customer_id)
group by ct.credit_card_type;
```

Output:

 Query 10

Query:

```
select
  ct.credit_card_type,
  round(avg(ot.discount),3) as average_discount,
  max(ot.discount) as max_discount,
  min(ot.discount) as min_discount
from customer_t ct
join order_t ot
  using(customer_id)
group by ct.credit_card_type
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	average_discount	max_discount	min_discount
americanexpress	0.616	0.8	0.4
bankcard	0.61	0.8	0.4
china-unionpay	0.622	0.78	0.4
diners-club-carte-blanc	0.614	0.76	0.42
diners-club-enroute	0.6	0.78	0.41
diners-club-international	0.584	0.63	0.55
diners-club-us-ca	0.615	0.76	0.45
instapayment	0.621	0.78	0.46

Observations and Insights:

- Overall, this table suggest that the company is giving massive discounts (over 50% in all cases). Additionally, the lowest discount offered was 40%, while the highest was 80%
- As an outsider, this leads me to question the business model: are they over-pricing their product to make it seem like the customer is getting a huge discount, when in reality they aren't? Or is the company giving money away?
- Ultimately, there may be reasons to provide these discounts; however, I question the need for the company to discount every order. Doing so can lead to the perception that 40% off is the actual base price, so a lack of a higher discount can be perceived as a lack of any discount at all.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
select
    quarter_number,
    round(avg(julianday(ship_date) - julianday(order_date)), 3) as avg_delivery_time
from order_t
group by quarter_number
order by quarter_number asc;
```

Output:

 Query 11

Query:

```
select
    quarter_number,
    round(avg(julianday(ship_date) - julianday(order_date)), 3) as avg_delivery_time
from order_t
group by quarter_number
order by quarter_number asc
```

Output:

Showing 4 rows

quarter_number	avg_delivery_time
1	57.168
2	71.111
3	117.755
4	174.095

Observations and Insights:

- Overall, the company has seen its delivery time more than triple to nearly half a calendar year.
- Any value a customer might gain from using the company's services as an end-to-end service provider could very well be lost by the customer having to wait nearly six months for their purchase.
- While the causes of this are not visible in the data, this trend seems to violate the core purpose of New Wheels' business model.

Business Metrics Overview

Total Net Revenue	Total Orders	Total Customers	Average Rating
\$48,610,993.78	1000	994	3.14
Last Quarter Net Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
\$8,573,149.28	199	97.96	44.10%

Key Insights

- The average customer rating is slightly higher than “Okay” at a 3.14 (scale is 1-5 from Very Bad to Very Good).
- Last quarter orders account for 20% of all orders rather than an expected 25% (assuming sales remain constant through the year).
- Last Quarter Net revenue accounted for only about 16% of annual revenue rather than an expected 25% assuming constant sales.
- There are only 6 repeat customers out of the 1000 orders in the data set.
- Less than half of all feedback during the year can be considered positive (Good or Very Good).

Business Recommendations

- The business must immediately focus on stopping the bleeding. Rapid measures should be implemented to improve several metrics across the board.
- First, average shipping times must be brought down. The Q4 shipping time is what is raising the overage to over 90 days. The company should ideally reduce shipping time to less than 60 days as seen in Q1. The company should immediately investigate its shipping partners (not part of this analysis although data is available) to see if one or more of its partners have led to the massive increase in days to ship.
- The goal of the business is to increase customer confidence in the end-to-end service of the vehicle resale market. This is evidently has not been occurring throughout this last year, as seen in the lower than 50% positive feedback. The company needs to launch a multi-tiered investigation in all departments to determine specific, actionable steps that can be taken to improve the customer experience. This should cover direct marketing, web design and customer facing app layout, shipping times, touch points, and customer tracking. Some action items should be rapid-deployed while others should be identified that require a culture shift. The company should set quarter over quarter goals for the improvement of its feedback scores. By the end of the year, the company should have an average of 50%+ customer satisfaction for the year with Q4 The satisfaction being at least what it

was in Q4 last year. Additionally, the goal should be to drop Very Bad rankings to as close to zero as possible.

- The company should also investigate its discount policy. This may need to include a complete pricing restructure. The company should bill itself as an affordable alternative to visiting a used-car reseller by offering standard lower prices across the board and offering specific discounts during promotions and sales events rather than having a “standard” 40% discount. The net pricing does not necessarily need to change, but the language and branding around discounts and pricing does.
- Finally, the company needs to invest in a data analyst department that specifically monitors these changes. The department should be responsible for the analysis and interpretation of all company data. The department should also design dashboards and other deliverables that allow all employees to track overall metrics, with drill down options and more detailed dashboards available to team leads.

All code for this project is pasted below.

```
-- Question 1:
-- Find the total number of customers who have placed orders. What is the distribution of the
customers across states? [4 marks]
-- Hint: For each state, count the number of customers

select
    count(distinct customer_id) as customers,
    ct.state
from customer_t ct
join order_t ot
    using(customer_id)
group by ct.state
    order by count(distinct customer_id) desc;

-- Question 2:
-- Which are the top 5 vehicle makers preferred by the customers? [4 marks]
-- Hint: For each vehicle make what is the count of the customers.

select
    pt.vehicle_maker,
    sum(ot.quantity) as total_units,
    round((sum(ot.quantity)*100.0/(select sum(quantity) from order_t)),2) as percent_market
from order_t ot
join product_t pt
    on ot.product_id = pt.product_id
group by pt.vehicle_maker
    order by total_units desc
limit 5;

-- Question 3:
-- Which is the most preferred vehicle maker in each state? [4 marks]
-- Hint: Use the window function RANK() to rank based on the count of customers for each state and
vehicle maker.
-- After ranking, take the vehicle maker whose rank is 1.

select
    state,
    vehicle_maker
from (
select
    ct.state,
    pt.vehicle_maker,
    count(distinct ct.customer_id) as total_customers,
    rank() over (partition by ct.state
        order by count(distinct ct.customer_id) desc) as rnk
from customer_t ct
join order_t ot
```

```

        using(customer_id)
join product_t pt
        using(product_id)
group by ct.state, pt.vehicle_maker) as state_ranks
where rnk = 1;

```

```

SELECT
    state,
    MIN(vehicle_maker) AS vehicle_maker,
    MAX(total_customers) AS total_customers
FROM (
    SELECT
        ct.state,
        pt.vehicle_maker,
        COUNT(DISTINCT ct.customer_id) AS total_customers,
        RANK() OVER (
            PARTITION BY ct.state
            ORDER BY COUNT(DISTINCT ct.customer_id) DESC
        ) AS rnk
    FROM customer_t ct
    JOIN order_t ot USING (customer_id)
    JOIN product_t pt USING (product_id)
    GROUP BY ct.state, pt.vehicle_maker
) AS state_ranks
WHERE rnk = 1
GROUP BY state
HAVING COUNT(*) = 1
order by total_customers desc;

```

```

-- Question 4:
-- Find the overall average rating given by the customers. What is the average rating in each
quarter? [5 marks]
-- Consider the following mapping for ratings:
-- "Very Bad": 1, "Bad": 2, "Okay": 3, "Good": 4, "Very Good": 5
-- Hint: Use subquery and assign numerical values to feedback categories using a CASE statement.
-- Then, calculate the average feedback count per quarter. Use a subquery to convert feedback
-- into numerical values and group by quarter_number to compute the average.

```

```

select
    quarter_number,
    round(avg(feedback_rating),3) as avg_rating,
    round(avg(avg(feedback_rating)) over(),3) as overall_avg,
    max(feedback_rating) as max_rating,
    sum(case when feedback_rating = 5 then 1 else 0 end) as max_count,
    min(feedback_rating) as min_rating,
    sum(case when feedback_rating = 1 then 1 else 0 end) as min_count
from (
    select
        quarter_number,
        case
            when customer_feedback = 'Very Bad' then 1
            when customer_feedback = 'Bad' then 2

```

```

        when customer_feedback = 'Okay' then 3
        when customer_feedback = 'Good' then 4
        when customer_feedback = 'Very Good' then 5
    end as feedback_rating
    from order_t)
group by quarter_number
order by quarter_number asc;

```

-- Question 5:

-- Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time? [5 marks]

-- Hint: Calculate the percentage of each feedback type by using conditional aggregation.

-- For each feedback category, use a CASE statement to count the occurrences and then divide by the total count of feedback for the quarter, multiplied by 100 to get the percentage.

-- Finally, group by quarter_number and order the results to reflect the correct sequence.

```

select
    quarter_number,
    round(sum(case when customer_feedback = 'Very Bad' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Very_Bad,
    round(sum(case when customer_feedback = 'Bad' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Bad,
    round(sum(case when customer_feedback = 'Okay' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Okay,
    round(sum(case when customer_feedback = 'Good' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Good,
    round(sum(case when customer_feedback = 'Very Good' then 1 else 0
        end)*100.0 / count(customer_feedback),2) as Very_Good
from order_t
group by quarter_number;

```

-- Question 6:

-- What is the trend of the number of orders by quarter? [3 marks]

-- Hint: Count the number of orders for each quarter.

```

select
    quarter_number as quarter,
    count(order_id) as total_orders,
    sum(quantity) as total_quant
from order_t
group by quarter_number;

```

-- Question 7:

-- Calculate the net revenue. What is the quarter-over-quarter % change? [5 marks]

-- Hint: Net Revenue is the amount obtained by multiplying the number of units sold by the price after deducting the discounts applied.

-- Quarter over Quarter percentage change in revenue means what is the change in revenue from the subsequent quarter to the previous quarter in percentage.

```
-- Calculate the revenue for each quarter by summing the quantity of product
and the discounted vehicle price. Use the LAG function to get the revenue
from the previous quarter, and then compute the quarter-over-quarter
percentage change based on the current and previous revenue values.
-- Ensure the results are ordered by quarter_number to maintain the correct sequence.
```

```
select
    quarter_number,
    net_revenue as current_q_revenue,
    lag(net_revenue) over (order by quarter_number) as previous_q_revenue,
    round((net_revenue - lag(net_revenue) over (order by quarter_number))
        *100.0 / lag(net_revenue) over (order by quarter_number), 2) as percent_change
from (
select
    quarter_number,
    round(sum((vehicle_price * (1-discount)) * quantity), 2) as net_revenue
from order_t
group by quarter_number
order by quarter_number asc);
```

```
-- Question 8:
-- What is the combined trend of net revenue and orders by quarter? [4 marks]
-- Hint: Find out the sum of net revenue and count the number of orders for each quarter.
```

```
select
    quarter_number,
    total_orders,
    round((total_orders - lag(total_orders) over(order by quarter_number))
        *100.0 / lag(total_orders) over(order by quarter_number), 2) as perc_chg_orders,
    net_revenue,
    round((net_revenue - lag(net_revenue) over (order by quarter_number))
        *100.0 / lag(net_revenue) over(order by quarter_number), 2) as perc_chg_rev
from (
select
    quarter_number,
    count(order_id) as total_orders,
    round(sum((vehicle_price * (1-discount)) * quantity), 2) as net_revenue
from order_t
group by quarter_number
order by quarter_number asc);
```

```
-- Question 9:
-- What is the average discount offered for different credit card types? [3 marks]
-- Hint: Find out the average of discount for each credit card type.
```

```
select
    ct.credit_card_type,
    round(avg(ot.discount),3) as average_discount,
```



```

max(ot.discount) as max_discount,
min(ot.discount) as min_discount
from customer_t ct
join order_t ot
    using(customer_id)
group by ct.credit_card_type;

-- Question 10:
-- What is the average time taken to ship the placed orders for each quarter? [3 marks]
-- Hint: Please use the julianday function instead of the DATEDIFF function to find the difference
between the ship date and the order date.
-- The SQL Playground Editor is built on the SQLite platform, which doesn't support the DATEDIFF
function available in MySQL

select
    quarter_number,
    round(avg(julianday(ship_date) - julianday(order_date)), 3) as avg_delivery_time
from order_t
group by quarter_number
order by quarter_number asc;

-- Summary Metrics

select
    round(sum((ot.vehicle_price * (1-ot.discount)) * quantity), 2) as total_revenue,
    count(ot.order_id) as total_orders,
    count(distinct ct.customer_id) as total_customers,
    round(avg(
        case
            when customer_feedback = 'Very Bad' then 1
            when customer_feedback = 'Bad' then 2
            when customer_feedback = 'Okay' then 3
            when customer_feedback = 'Good' then 4
            when customer_feedback = 'Very Good' then 5
        end), 2) as avg_feedback_rating,
    round(sum(case when ot.quarter_number = 4 then
        (ot.vehicle_price * (1-ot.discount)) * quantity end),2) as q4_rev,
    count(case when ot.quarter_number = 4 then ot.order_id end) as q4_orders,
    round(avg(julianday(ot.ship_date) - julianday(ot.order_date)), 2) as avg_delivery_time,
    round((count(case when ot.customer_feedback in ('Good', 'Very Good') then 1 end)
        *100.0 / count(ot.customer_feedback)), 2) as good_feedback
from customer_t ct
join order_t ot
    using(customer_id)

```

